

## BROOKINGS, SD, USA

WMO: 726515

Lat: 44.300N

Lon: 96.800W

Elev: 1648

StdP: 13.84

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94902

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -14.6      | -8.9 | -25.9                         | 1.4 | -8.0 | -19.8 | 2.0 | -6.5 | 31.9                  | 12.9 | 28.1 | 13.5 | 7.7                   | 320  | 0.641 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.9                   | 89.6            | 73.7 | 86.0 | 71.8 | 82.8 | 69.8 | 75.9                | 85.6 | 74.1 | 83.1 | 72.2 | 80.5 | 13.4                 | 190  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 72.7                            | 129.1 | 82.1 | 71.6 | 123.9 | 80.9 | 69.6 | 115.9 | 79.2 | 40.6          | 86.2 | 38.8 | 82.7 | 36.9 | 80.6 | 82.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |    |    |
| (4) 27.0          | 24.3 | 20.9 | -21.8                      | 94.3 | 5.6                | 2.7 | -25.8                                              | 96.3 | -29.1      | 97.9 | -32.2      | 99.4 | -36.3      | 101.4 |    |    |
| (5)               |      |      | -20.6                      | 78.8 | 5.0                | 1.9 | -24.3                                              | 80.2 | -27.2      | 81.3 | -30.0      | 82.4 | -33.7      | 83.8  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 44.3   | 14.1  | 18.8  | 30.4  | 44.0  | 56.4  | 66.5  | 71.6  | 68.4  | 60.6  | 47.0  | 32.6  | 19.4  |  |  |
|                                                                        | DBStd       | 22.31  | 13.00 | 13.16 | 12.92 | 10.17 | 8.71  | 6.92  | 5.94  | 5.79  | 8.35  | 9.58  | 10.80 | 11.95 |  |  |
|                                                                        | HDD50       | 4530   | 1113  | 874   | 616   | 232   | 37    | 0     | 0     | 0     | 12    | 171   | 526   | 949   |  |  |
|                                                                        | HDD65       | 8105   | 1578  | 1294  | 1072  | 633   | 292   | 63    | 14    | 32    | 180   | 562   | 971   | 1414  |  |  |
|                                                                        | CDD50       | 2446   | 0     | 0     | 9     | 51    | 235   | 496   | 670   | 570   | 332   | 78    | 5     | 0     |  |  |
|                                                                        | CDD65       | 543    | 0     | 0     | 0     | 2     | 24    | 108   | 219   | 137   | 49    | 4     | 0     | 0     |  |  |
|                                                                        | CDH74       | 4943   | 0     | 0     | 3     | 39    | 288   | 1000  | 2025  | 1096  | 439   | 53    | 0     | 0     |  |  |
|                                                                        | CDH80       | 1435   | 0     | 0     | 0     | 8     | 80    | 288   | 668   | 285   | 98    | 8     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 9.9    | 9.9   | 10.4  | 10.9  | 11.8  | 10.8  | 9.1   | 8.4   | 8.2   | 9.4   | 10.3  | 10.0  | 9.8   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 22.8   | 0.3   | 0.4   | 1.2   | 2.1   | 2.8   | 4.2   | 3.2   | 2.9   | 2.6   | 1.6   | 0.8   | 0.4   |  |  |
|                                                                        | PrecMax     | 32.3   | 1.2   | 2.4   | 3.5   | 5.4   | 9.3   | 9.3   | 12.1  | 7.0   | 7.7   | 6.4   | 4.4   | 2.1   |  |  |
|                                                                        | PrecMin     | 13.3   | 0.0   | 0.0   | 0.2   | 0.3   | 0.4   | 0.8   | 0.8   | 0.9   | 0.1   | 0.0   | 0.0   | 0.0   |  |  |
|                                                                        | PrecStd     | 5.0    | 0.4   | 0.5   | 0.9   | 1.2   | 2.0   | 2.2   | 2.4   | 1.5   | 1.7   | 1.5   | 0.9   | 0.4   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 45.4  | 55.2  | 69.9  | 79.5  | 88.0  | 91.0  | 93.4  | 90.6  | 87.9  | 81.0  | 66.3  | 50.0  |  |  |
|                                                                        |             | MCWB   | 38.5  | 46.9  | 56.0  | 60.0  | 67.0  | 73.0  | 75.4  | 74.6  | 70.7  | 62.5  | 52.9  | 41.4  |  |  |
|                                                                        | 2%          | DB     | 37.4  | 45.7  | 61.2  | 72.2  | 81.4  | 86.5  | 90.1  | 86.3  | 82.3  | 72.7  | 60.5  | 43.3  |  |  |
|                                                                        |             | MCWB   | 33.6  | 39.4  | 51.9  | 55.4  | 63.6  | 70.8  | 75.1  | 73.3  | 68.3  | 59.5  | 49.5  | 37.1  |  |  |
|                                                                        | 5%          | DB     | 35.3  | 39.9  | 54.7  | 65.9  | 76.3  | 83.0  | 87.6  | 82.4  | 79.4  | 66.5  | 54.0  | 37.4  |  |  |
|                                                                        |             | MCWB   | 32.8  | 35.6  | 47.6  | 52.2  | 61.1  | 69.2  | 73.7  | 70.5  | 66.0  | 55.5  | 45.6  | 33.4  |  |  |
|                                                                        | 10%         | DB     | 31.9  | 36.4  | 47.9  | 61.2  | 72.1  | 80.5  | 83.7  | 80.7  | 75.2  | 63.0  | 47.8  | 35.0  |  |  |
|                                                                        |             | MCWB   | 30.3  | 33.0  | 42.6  | 49.1  | 58.8  | 66.9  | 71.6  | 69.0  | 64.4  | 53.8  | 41.3  | 32.1  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 38.9  | 47.7  | 58.4  | 61.5  | 69.7  | 76.3  | 78.7  | 77.3  | 74.0  | 66.4  | 55.5  | 42.7  |  |  |
|                                                                        |             | MCDB   | 45.2  | 56.2  | 65.0  | 75.7  | 83.2  | 86.5  | 88.4  | 86.8  | 81.9  | 75.8  | 63.9  | 48.5  |  |  |
|                                                                        | 2%          | WB     | 34.6  | 39.6  | 52.8  | 57.2  | 66.1  | 73.4  | 76.6  | 75.0  | 71.1  | 61.5  | 50.2  | 37.3  |  |  |
|                                                                        |             | MCDB   | 37.4  | 44.9  | 59.3  | 69.2  | 78.0  | 83.4  | 87.0  | 83.5  | 79.1  | 69.6  | 58.3  | 42.5  |  |  |
|                                                                        | 5%          | WB     | 32.5  | 36.3  | 47.9  | 53.7  | 63.3  | 70.7  | 74.9  | 73.0  | 68.9  | 57.9  | 46.0  | 34.2  |  |  |
|                                                                        |             | MCDB   | 34.9  | 40.6  | 54.8  | 63.0  | 73.4  | 81.0  | 84.4  | 80.8  | 77.2  | 66.1  | 53.4  | 37.5  |  |  |
|                                                                        | 10%         | WB     | 30.0  | 32.9  | 42.2  | 50.7  | 60.2  | 68.2  | 73.0  | 70.8  | 66.4  | 54.1  | 41.7  | 32.1  |  |  |
|                                                                        |             | MCDB   | 31.8  | 36.1  | 47.8  | 59.7  | 69.4  | 78.5  | 81.7  | 78.0  | 73.6  | 61.3  | 47.4  | 34.6  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 16.5  | 17.5  | 17.1  | 20.0  | 20.1  | 19.4  | 19.9  | 19.6  | 20.9  | 19.9  | 17.9  | 15.8  |  |  |
|                                                                        |             | MCDBR  | 19.2  | 21.7  | 26.2  | 29.6  | 27.2  | 23.3  | 22.4  | 22.0  | 25.0  | 28.3  | 27.5  | 20.3  |  |  |
|                                                                        |             | MCWBR  | 15.9  | 16.5  | 17.6  | 16.8  | 14.7  | 12.3  | 11.6  | 12.3  | 13.5  | 17.1  | 18.1  | 15.7  |  |  |
|                                                                        | 5% WB       | MCDBR  | 18.3  | 20.9  | 25.2  | 25.8  | 23.5  | 21.1  | 20.1  | 19.7  | 20.9  | 24.0  | 26.2  | 19.4  |  |  |
|                                                                        |             | MCWBR  | 15.3  | 16.1  | 17.3  | 16.1  | 13.9  | 12.4  | 11.7  | 12.2  | 12.7  | 15.6  | 18.0  | 15.4  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.263 | 0.286 | 0.315 | 0.364 | 0.378 | 0.388 | 0.395 | 0.398 | 0.352 | 0.321 | 0.289 | 0.267 |  |  |
|                                                                        | taud        |        | 2.437 | 2.394 | 2.404 | 2.298 | 2.339 | 2.367 | 2.343 | 2.315 | 2.420 | 2.483 | 2.482 | 2.442 |  |  |
|                                                                        | Ebn at Noon |        | 278   | 289   | 293   | 285   | 283   | 279   | 276   | 271   | 276   | 272   | 265   | 264   |  |  |
|                                                                        | Edh at Noon |        | 23    | 28    | 32    | 39    | 39    | 38    | 38    | 38    | 32    | 26    | 22    | 20    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 591   | 924   | 1204  | 1526  | 1773  | 1976  | 2119  | 1749  | 1377  | 914   | 616   | 455   |  |  |
|                                                                        | RadStd      |        | 62    | 90    | 118   | 146   | 132   | 136   | 76    | 143   | 104   | 111   | 55    | 39    |  |  |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (38 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page